

[54] POSITIONING TYPEWRITER

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[51] Int. Cl. **B41j 5/30**

[58] Field of Search 197/18, 19, 50, 20, 16,
197/55, 1 R, 48; 101/93 C; 340/172.5

[56] **References Cited**

UNITED STATES PATENTS

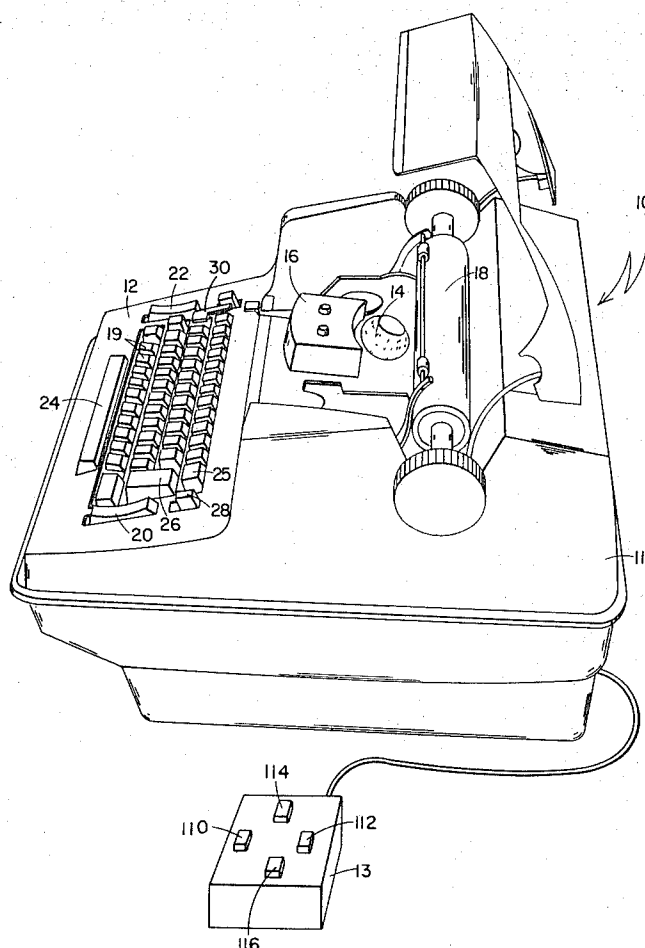
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Primary Examiner—Robert E. Pulfrey
Assistant Examiner—R. T. Rader

[57] **ABSTRACT**

A writer selectively operable in a alpha-numeric symbol printing mode and a positioning mode. The writer includes a horizontally movable type head having a plurality of corresponding alpha-numeric printing symbols, operable in response to printing signals to print the symbols and horizontally movable in response to horizontal spacing function signals; and a platen rotatable in response to vertical spacing function signals. Type head drive means are provided for incrementally horizontally moving the type head in fine increments, the alpha-numeric spacing increments being an integral multiple thereof, and platen drive means are provided for incrementally rotating the platen in fine increments. In general, this is done by control means for operating the type head comprising horizontal function signal generating means for incrementally horizontally moving the type head in fine increments, and platen drive means comprising vertical function signal generating means for incrementally rotating the platen in fine increments, the function keys providing function signals being connected to the function signal generating means for generating a plurality of fine increment signals equal to an alpha-numeric increment.

6 Claims, 11 Drawing Figures



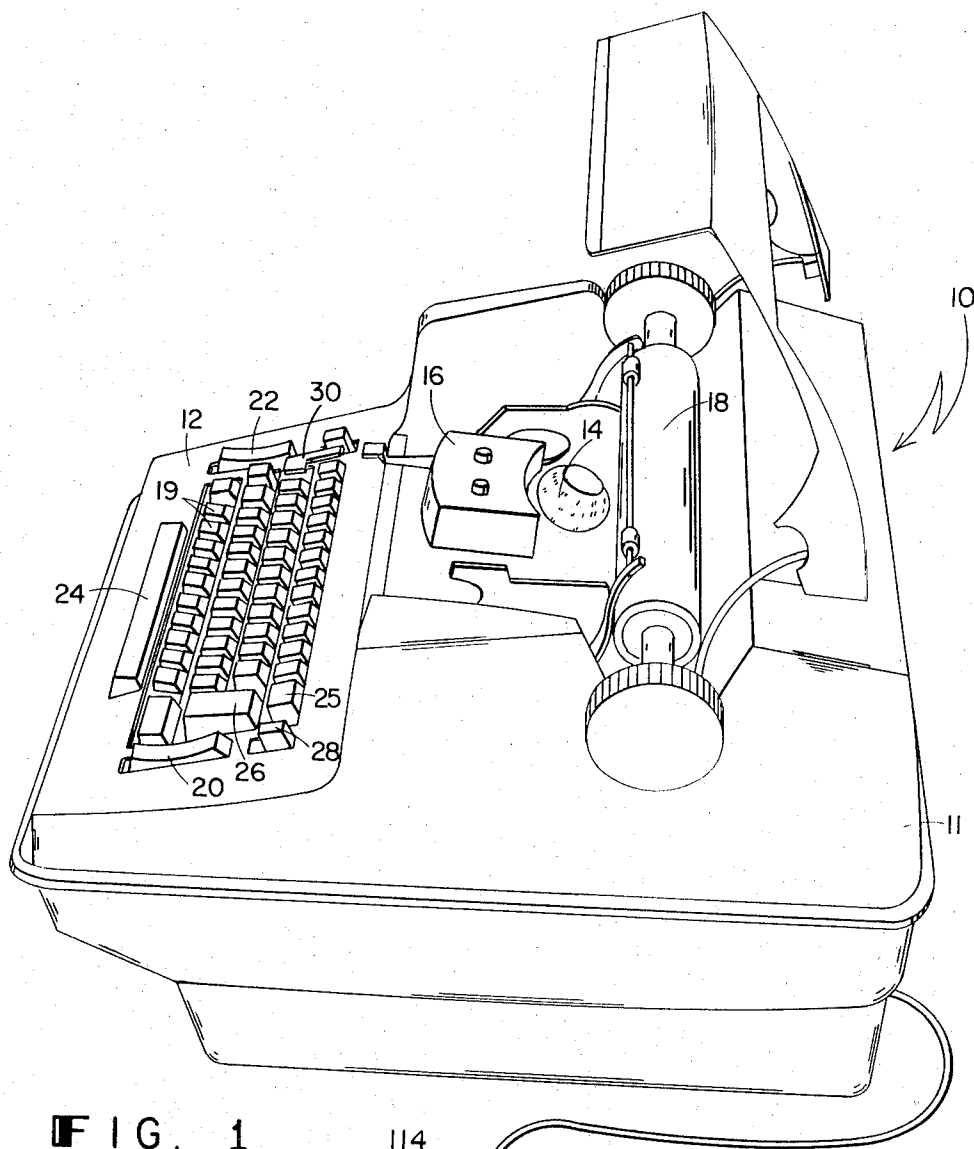
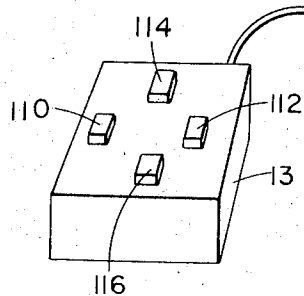


FIG. 1



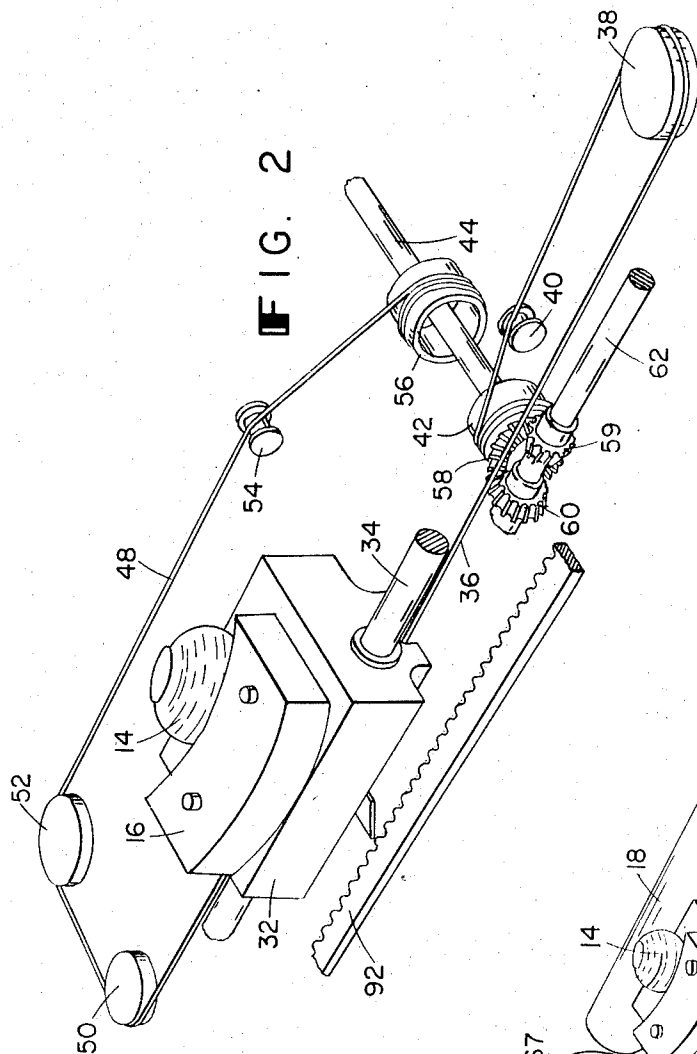


FIG. 2

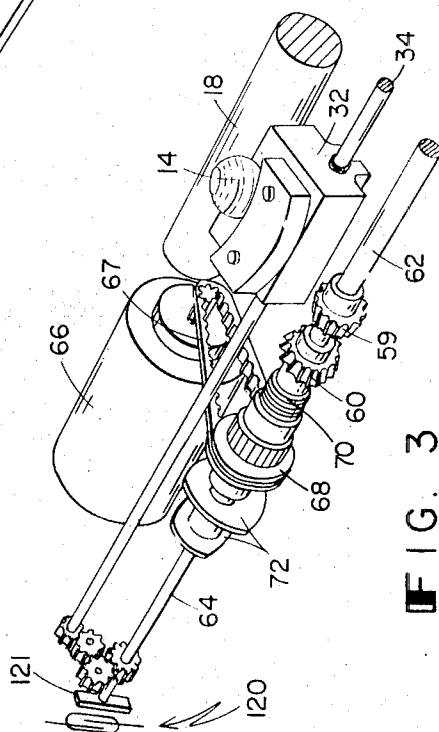


FIG. 3

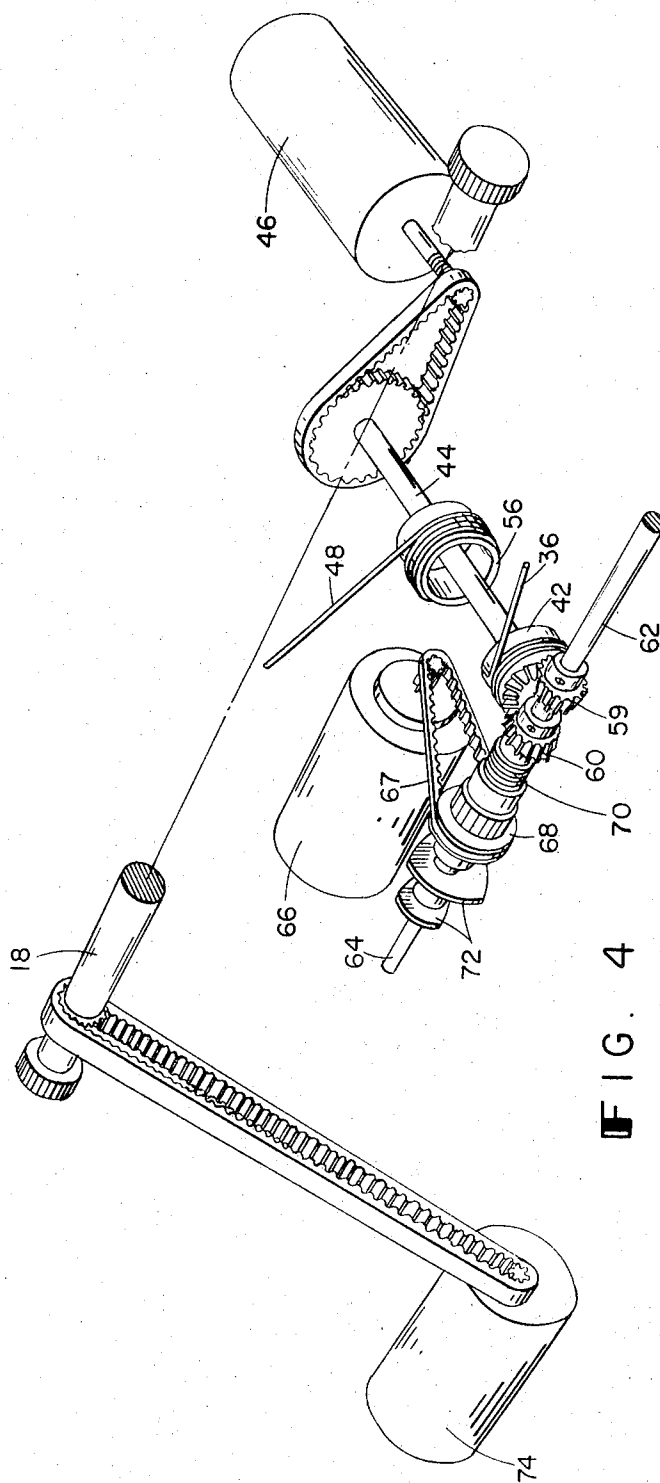


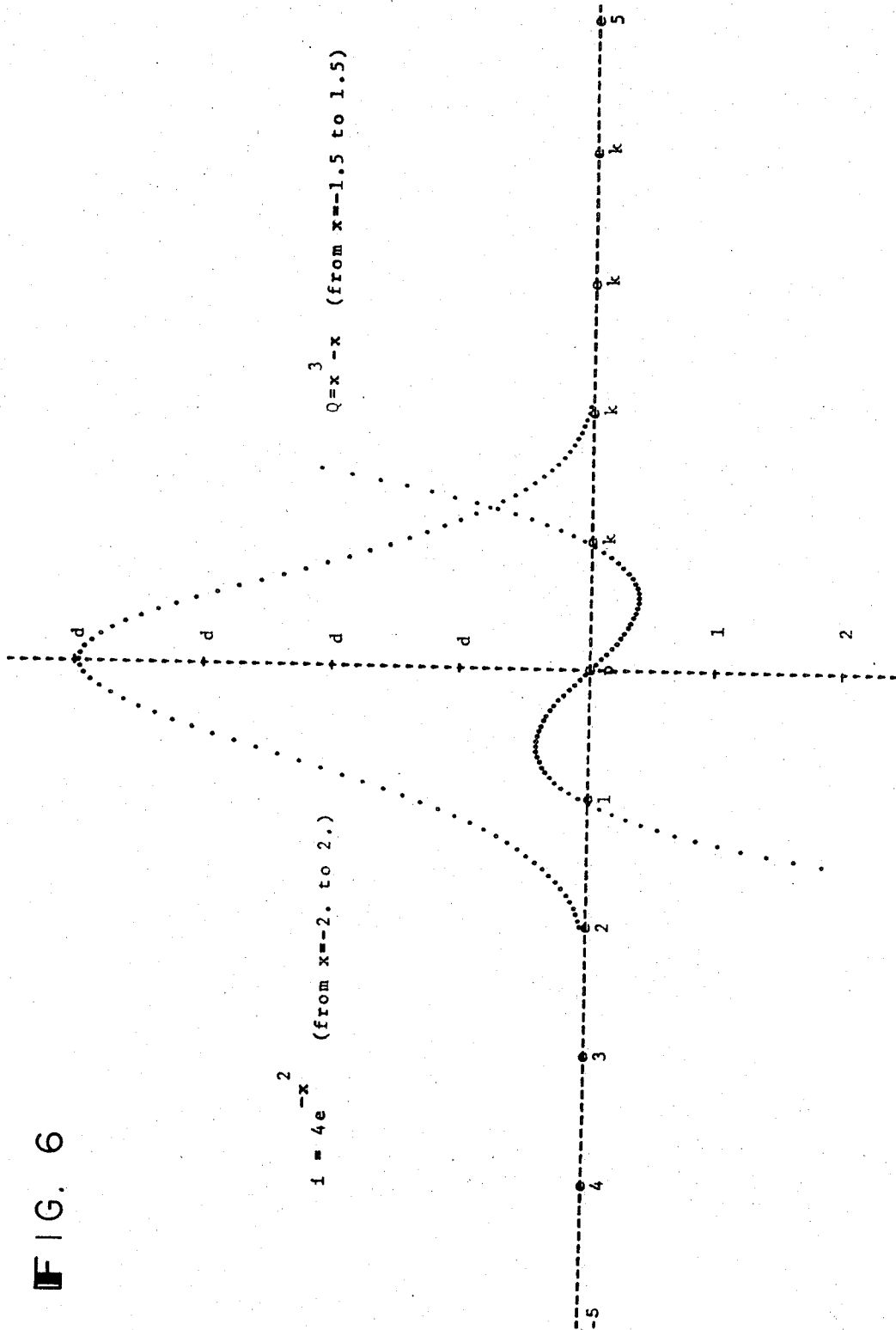
FIG 5

CODES FOR PRINTING AND PLOTTING OUTPUT-WRITERS

	(Rotation)				Home Position				T ₁ KA ₀ (Tilt) T ₂ KA ₁
	R ₁ KB ₀	R ₂ KB ₁	R _{2A} KB ₂	R ₅ KB ₃	1	0	1	0	
PRINT (LC)					1	0	1	0	
					1	1	0	0	
	1	1	1	1	1	x	m	g	
	0	1	1	1	3	u	v	f	
	1	0	1	1	7	d	r	;	
	0	0	1	1	8	c	a	,	
	1	0	0	1	4	l	o	/	
	1	1	1	0	z	t	½	j	Home Position
	0	1	1	0	2	n	.	=	
	1	0	1	0	5	e	'	p	
	0	0	1	0	6	k	i	q	
	1	0	0	0	0	h	s	y	
	0	0	0	0	9	b	w	-	
FUNCTION PLOTTING & PRINTING	0	0	0	1	PLOT MODE	PRINT MODE	RETURN*		
	1	1	0	1	STEP X-Y-	STEP Y-	REV INDEX		
	0	1	0	1	" X-Y-	" Y+	INDEX		
	1	1	0	0	" X-Y+	" X-	SHIFT UP	HK SPACE	
	0	1	0	0	" X+Y+	" X+	SHIFT DN	SPACE	

* Return will be automatically followed by an index operation in the PRINT MODE

FIG. 6



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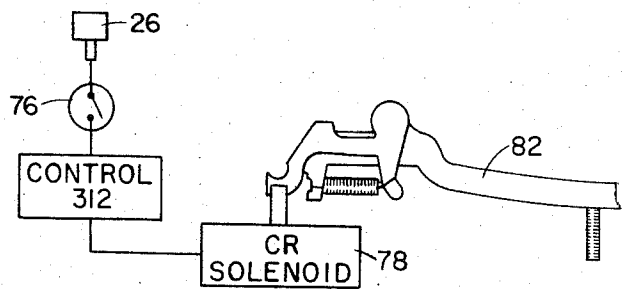


FIG. 7

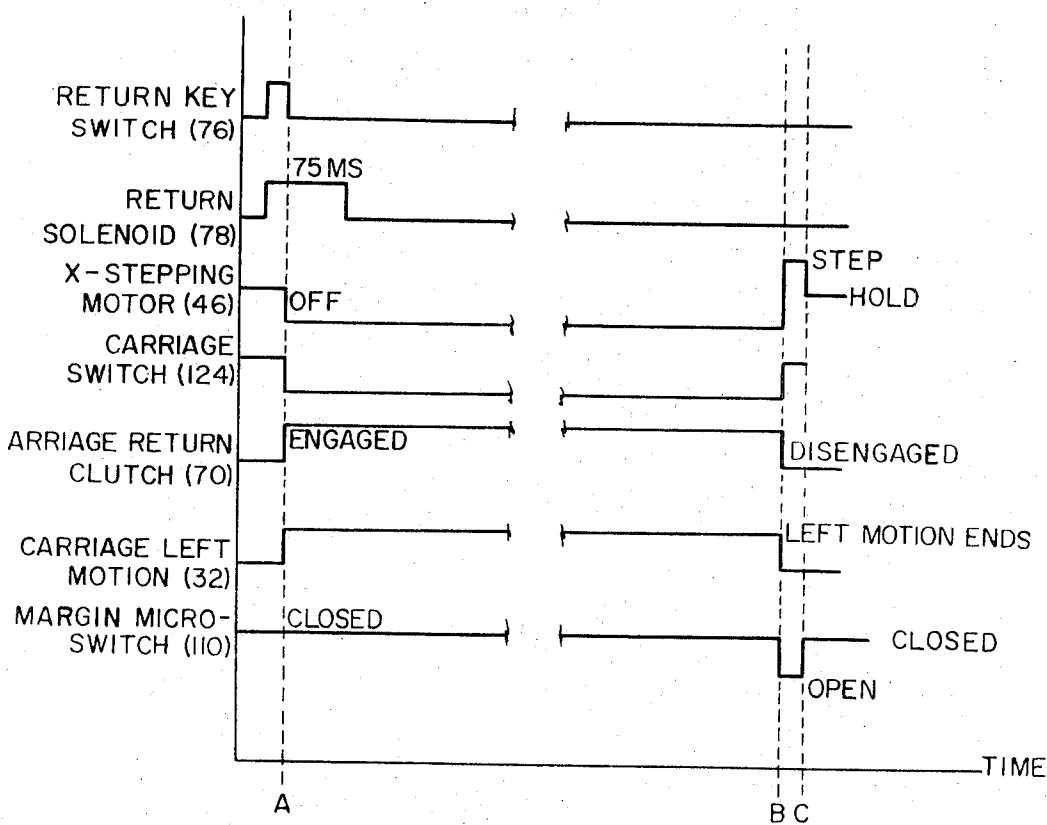
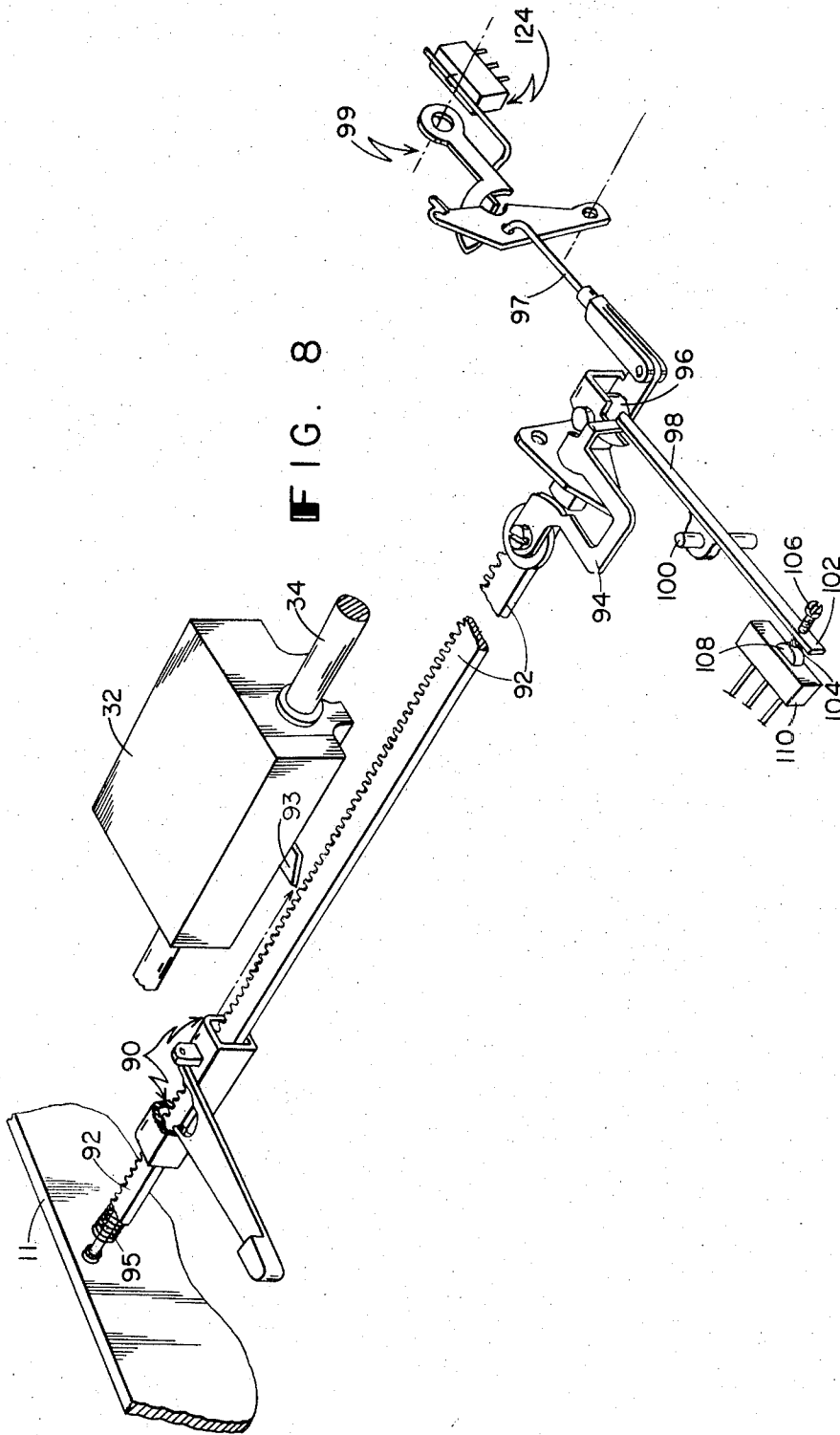


FIG. 9



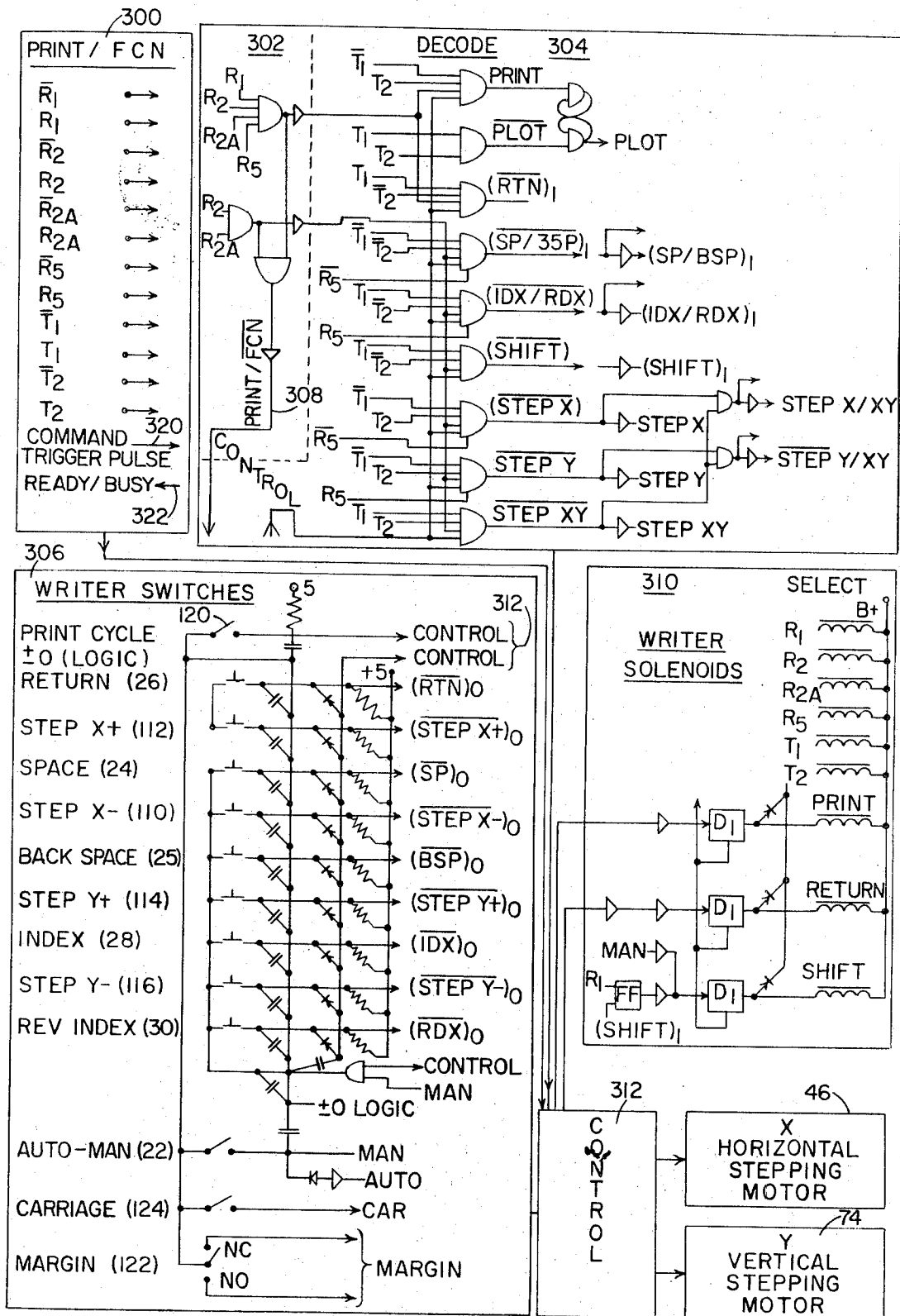


FIG. 10

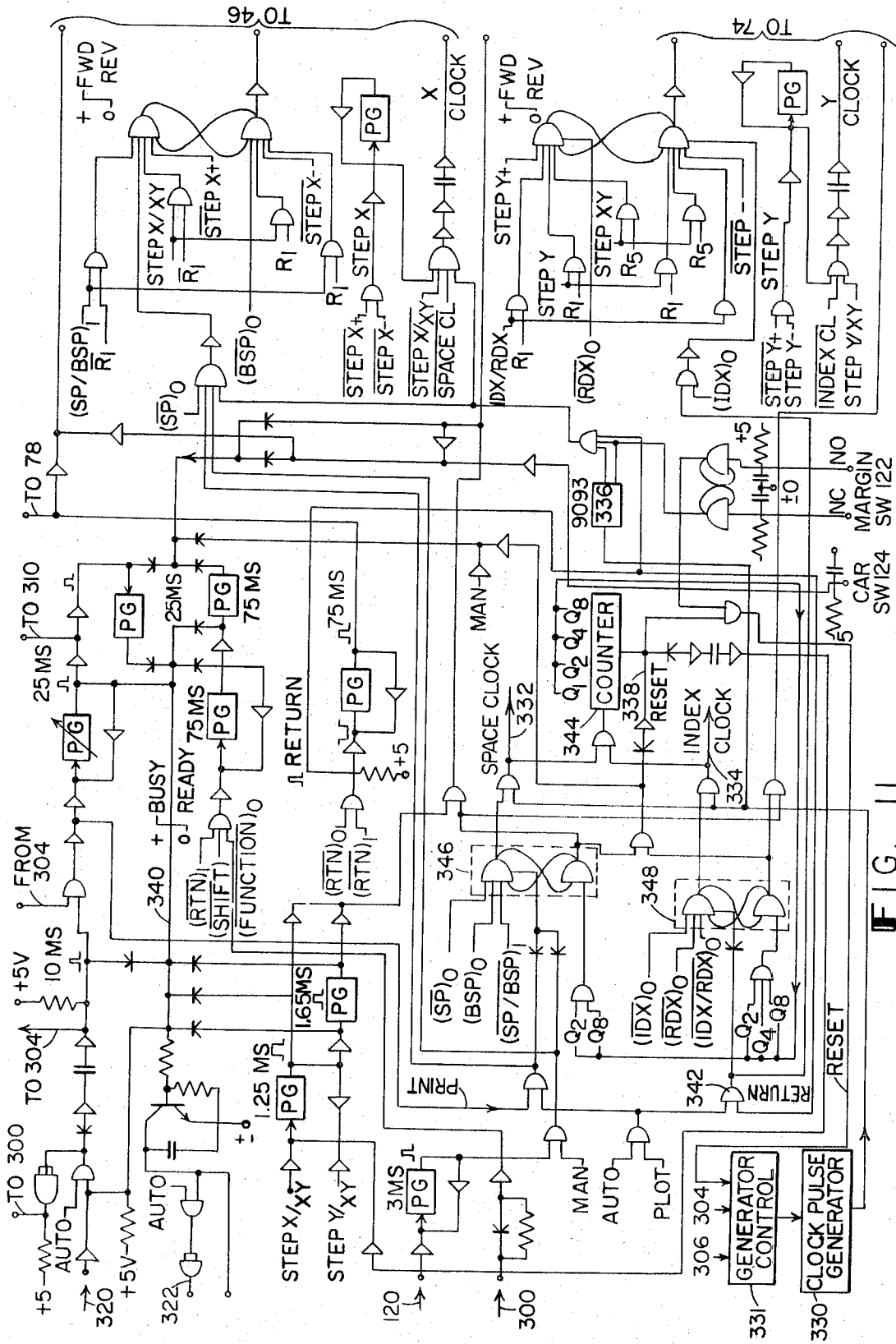


FIG. 11

POSITIONING TYPEWRITER

This application is a division of our application Ser. No. 130, 281, filed Apr. 1, 1971.

This invention relates to typewriters and more particularly to those used as output writers for computers and the like.

The conventional computer input/output typewriter, of the IBM Selectric type, for example, although widely used, has a number of significant limitations. Among the most important of these is its inability to position its symbols relatively to one another in other than alpha-numeric spacing increments, which greatly restricts its effectiveness as a plotter, as well as its use for composition, justification or the like. Other such limitations are related to the speed of operation of the IBM typewriter, and include, for example, its practical inability to type reversibly both horizontally and vertically, its inability simultaneously to space horizontally and vertically, and its inability to space during the preliminary stages of a printing stroke.

Accordingly, it is a major object of the present invention to provide a novel positioning typewriter movable in fine increments, much less than an alpha-numeric spacing increment.

It is another object of the invention to provide such a typewriter while retaining its conventional operation, including manual operation.

It is still another object of the invention to provide such a typewriter by novel modifications to a typewriter of the IBM Selectric type.

Further objects of the invention are the provision of a typewriter capable of typing reversibly both horizontally and vertically, of simultaneously spacing both vertically and horizontally, and of spacing during the preliminary stages of a printing stroke.

The above and still further objects and features of the present invention are uniquely accomplished by providing, in a major aspect of the invention, a writer selectively operable in an alpha-numeric symbol printing mode and a positioning mode. Such writer includes a keyboard with a plurality of symbol printing keys providing printing signal and horizontal spacing function signals representing alpha-numeric spacing increments and a plurality of function keys providing horizontal and vertical function signals representing alpha-numeric spacing increments, including horizontal space and vertical index keys; a horizontally movable type head having a plurality of corresponding alpha-numeric printing symbols, operable in response to the printing signals to print the symbols and horizontally movable in response to the horizontal spacing function signals; and a platen rotatable in response to the vertical spacing function signals.

A major characteristic unique to the invention is its unique provision of type head drive means for incrementally horizontally moving the type head in fine increments, the alpha-numeric spacing increments being an integral multiple thereof, and platen drive means for incrementally rotating the platen in fine increments. In general, this is done by control means for operating the type head comprising horizontal function signal generating means for incrementally horizontally moving the type head in fine increments, and platen drive means comprising vertical function signal generating means for incrementally rotating the platen in fine increments, the function keys providing function signals being connected to the function signal generating means for gen-

erating a plurality of fine increment signals equal to an alpha-numeric increment.

Other objects, features, and advantages will appear from the following description of a preferred embodiment of the invention, taken together with the attached drawings thereof, in which:

FIG. 1 is a perspective view of the exterior of the typewriter;

FIG. 2 is a perspective view of a portion of the typewriter, including a portion of the type head horizontal drive means;

FIG. 3 is a perspective view of a further portion of the type head horizontal drive means;

FIG. 4 is a perspective view including the platen drive means and parts of the type head horizontal drive means;

FIG. 5 is a table of codes for generating printing signals and function signals;

FIG. 6 is a typical typed product of the typewriter using fine increment spacing;

FIG. 7 is a schematic detailed portion of the type head horizontal drive means;

FIG. 8 is a perspective view of a portion of the type head horizontal drive means including left margin locating means;

FIG. 9 is a timing diagram representing return to the left margin;

FIGS. 10 and 11 represent the logic circuitry of the typewriter.

In the preferred embodiment described herein and shown in the drawings, most generally in FIG. 1, the inventive principles have been applied to an electric typewriter of the type designated Selectric, manufactured by International Business Machines, and described, for example, in U.S. Pat. No. 2,879,876, in the "IBM Selectric Instruction Manual" dated Jan., 1966, and in the "IBM Customer Engineering Manual of Instruction, Selectric I/O Keyboard Printer."

In an unmodified typewriter, a single print operation comprises the steps of selecting an alpha-numeric printing symbol, tilting and rotating the type head, striking the paper with the head, returning the head to rest position, and translating the head horizontally through a standard spacing increment to the next printing space. In the present embodiment, the mechanics of symbol selection, head tilting and rotation, striking and return are unaltered, and will not be described here. The horizontal motion of the type head to the next print space, however (as well as the vertical indexing), is accomplished in a novel manner in the present embodiment, as will be described.

Modifications of the horizontal spacing and vertical indexing elements unique to the present invention enable the typewriter to accomplish accurate fine positioning of a printing symbol, preferably with a minimum step of 0.02 inch, or more desirably 0.01 inch, in both horizontal and vertical spacing operations. At the same time, the typewriter retains the capability of spacing in standard increments when desired, such standard increments being integral multiples of the fine increments.

To accomplish this, the standard vertical and horizontal spacing operations have been modified by disabling the original mechanical linkages, and space rack detent, the mainspring, and platen drive, and removing the tabulation mechanism. Two stepping motors, of the type designated SLO-SYM and manufactured by the

Superior Electric Co. of Bristol, Connecticut, have been added to drive the type head horizontally and to rotate the platen, either in sequence or simultaneously, and either in fine increments or in standard alphanumeric spacing and indexing increments as desired. The keyboard has been modified and an auxiliary keyboard added to provide additional space function control keys for these motions, replacing other keys found on the unmodified typewriter.

For use as an output device, the typewriter of this invention has been further modified, by the addition of solenoids and logic circuitry, similarly to a Selectric Input/Output Keyboard Printer.

These modifications cause the typewriter of the invention to space only in response to electrical spacing signals generated by manual depression of spacing function keys on the keyboard, or, when desired, derived from codes input from an external source, and to print symbols either in response to depression of a symbol printing key, or to electrical printing signals derived from input codes.

Referring now to FIG. 1, the typewriter of the invention, generally designated 10, is contained in a housing 11 and provides a keyboard 12, a type head 14 and an associated ribbon mechanism 16, and a rotatable platen 18. Keyboard 12 provides a plurality of alphanumeric symbol printing keys 19, including punctuation symbol keys. Power to typewriter 10 is controlled by on/off switch 20, while the source of operation control signals (keyboard or other) is determined by manual/automatic switch 22 (replacing the tab clear/set key). Horizontal motion of type head 14 may be manually controlled by spacebar 24 and back space key 25, together with return key 26. Vertical motion of the paper (indexing) results from rotation of platen 18, and may be manually controlled by index key 28, reverse index key 30 (replacing the tab key), and return key 26 (since each return to the left margin is accompanied by a vertical shift or index). Additional controls mounted on auxiliary keyboard 13 for either manual or automatic operation may be provided in the form of fine increment keys 110 (step left, or $x-$), 112 (step right, or $x+$), 114 (step up, or $y+$) and 116 (step down, or $y-$), each key determining a single fine increment step in the indicated direction.

Referring now to FIG. 2, type head 14 is carried on a carriage, shown schematically at 32, which slides horizontally on rotatable print shaft 34. Carriage drive means are provided, including escapement cord 36, attached to carriage 32, which passes around pulley 38 through a tensioning device (not shown) to guide roller 40, and is wound onto escapement drum 42. Drum 42 is rotated by escapement shaft 44. In an unmodified typewriter, a rotary force tending to wind up cord 36 onto drum 42 is supplied to shaft 44 by a mainspring. In the present embodiment, the mainspring has been removed, and shaft 44 instead is connected through a pulley to the horizontal (x) axis stepping motor 46 (FIG. 4). Carriage return cord 48, attached to carriage 32, passes around pulleys 50 and 52 and guide roller 54 and is wound onto carriage return cord drum 56, which is rotated by shaft 44.

Referring to FIG. 3, drum 42 provides a bevel gear 58, adapted to mesh with a tab governor pinion 59 (disabled in the present embodiment) and a carriage return pinion 60. Main drive motor 66 through belt 67 drives the cycle clutch pulley 68 in the conventional manner.

Extending to the right from cycle clutch pulley 68 is the operational shaft 62, rotated by motor 66; shaft 62 in an unmodified typewriter drives the various spacing functions that are powered by stepping motors in the present embodiment. Extending to the left of pulley 68 is the cycle shaft 64, which powers the rotation and tilt of type head 14 to the desired print symbol through cams 72. Pulleys 68 rotates continuously whenever motor 66 is running, but cycle shaft 64 rotates only during a print operation. A cycle clutch spring of the usual type (not shown) is used to engage and disengage shaft 64 with pulley 68. Through gears 63, cycle shaft 64 turns print shaft 34 (FIG. 3) on which carrier 32 rides, to operate the print and ribbon mechanisms. A print cycle reed switch 120 is operated by a magnet 121 carried on the end of cycle shaft 64 and is actuated after each 180° of rotation of shaft 64.

Platen 18 drive means are provided by vertical (y) axis stepping motor 74 (FIG. 4).

Stepping motors of the type of motors 46 and 74 normally operate in two modes when energized; in one mode, the shaft rotates through a specified number of steps, providing an accurately defined displacement of the device whose motion is powered by the motor; in the second mode, the shaft is maintained at a fixed orientation to provide an accurate fixed position of the powered device. When the motor is not energized, the shaft rotates freely.

The finest practical increment for spacing by the stepping motors has been found to be about 0.01 to 0.02 inch. Finer increments have been found to be impractical, in part because most paper (except for specially prepared paper) is unstable under changes of humidity, and the dimensional alterations of the paper may be so great as to render plotting to finer accuracy useless. Also, increments of such dimensions are not readily resolved by the unaided eye, so that a series of periods, for example, appears as a continuous line.

Using the carriage and platen drive means described, when the modified typewriter is used in the alphanumeric symbol printing mode to provide standard typed material, a signal from switch 120 is sent to horizontal stepping motor through logic circuitry to be described, and in response, the shaft of stepping motor 46 rotates through 10 steps to power the motion of carriage 32 to the right through 10 fine increments of 0.01 inch, which is equivalent to the standard typing space used with pica type. Depression of space bar 24 or backspace key 25 on keyboard 12 similarly provides an appropriate signal to stepping motor 46, which powers horizontal motion of carriage 32 through 10 fine increments in the indicated direction.

Rotation of the platen through 14 fine increments, providing a standard line index space, is powered by vertical stepping motor 74 in response to a signal generated by depression of the index key 28, reverse index key 30 or return key 26 on key-board 12.

However, for justification of typed lines, or for production of accurately plotted graphical material, both horizontal and vertical spacing may be modified by the use of the keys on auxiliary keyboard 13. For example, depression of x -key 110 generates a signal which is transmitted to horizontal stepping motor 46, in response to which the stepping motor rotates through a single step, to power the horizontal motion of carriage 32 through one fine increment to the left. Thus the position at which the next alpha-numeric symbol will be

printed is only 9 fine increments away from the previous symbol, rather than a standard 10. Signals from the other fine increment keys similarly determine motion of the carriage through single fine increments to the right, and motion of the platen through single fine increments either up or down. A typical plot, typed by the typewriter using fine increment spacing, is shown in FIG. 6.

While x-axis stepping motor 46 may be used to drive type head carriage 32 horizontally either in a positive or in a negative sense with equal ease, during the usual alpha-numeric symbol printing operation, characters are printed only during left to right motion of carriage 32. When a complete line has been printed, type head carriage 32 must return to the left margin before the next line can be printed. Because of the operating speeds of the stepping motors used in this modified typewriter, the maximum stepping rate is 300 steps or 3 inches per second. To wait while the carriage returns at this rate is undesirable.

Therefore, the fast carriage return capability of a standard unmodified typewriter has been retained, although the mechanical control has been replaced by an electrical control, permitting this fast return function to be initiated in response to a code while the typewriter is controlled by an external device, as well as in response to depression of the return key, and permitting generation of an electrical signal to initiate automatic indexing. A switch and solenoid have been substituted for the mechanical linkage to the return key.

Referring now to FIG. 7, the mechanical linkage leading from Return key 26 in an unmodified typewriter has been disabled, and a return key switch 76 supplied, which is closed by depression of return key 26. Switch 76 actuates solenoid 78, which by pulling down on interposer 82 engages carriage return clutch 70 in the conventional manner. As clutch 70 is engaged, clutch latch 99 moves downward conventionally and thereby closes carriage switch 124, which causes x-axis stepping motor 46 to be de-energized for free rotation during the return. This sequence of events is shown at time point A in timing diagram of FIG. 9.

Since clutch 70 is engaged, operational shaft 62, turned by main motor 66, rotates carriage return pinion 60, which engages and rotates bevel gear 58 on drum 42. Rotation of drum 42 causes escapement shaft 44 to rotate in a direction to wind up carriage return cord 48 onto drum 56, while cord 36 is paid out from drum 42. Carriage 32 is moved to the left at the speed determined by main motor 66 (about 17 inch/sec).

Referring now to FIG. 8, margin rack 92 is mounted within housing 11 just in front of carriage 32. The left margin position of carriage 32 is coarsely determined mechanically by impact of margin stop latch 93 on left margin stop 90. In an unmodified typewriter, the exact left margin position is determined by a mechanical detent involving a pawl which engages a tooth in margin rack 92. However, in the present typewriter, that mechanism has been disabled to allow the use of stepping motors; therefore the precise left margin position must be determined in a different manner.

Rack 92 has a small amount of lateral play, and its left end is mounted through a spring 95; at most times, when carriage 32 is away from the left margin, spring 95 loads rack 92 to the right. During a fast return, when the carriage reaches the left margin, stop latch 93 on carriage 32 strikes left margin stop 90, impelling rack

92 to the left. The right end of rack 92 is attached by a U-shaped extension member 94 to a carriage return unlatching bell crank 96. When rack 92 moves to the left, bellcrank 96 pulls clutch release link 97 to disengage clutch 70 in the conventional manner, and at the same time the resultant motion of the carriage return clutch latch 99 opens carriage switch 124 to energize stepping motor 46.

Extension member 94 is also connected to a lever 98, which pivots about pin 100; end 102 of lever 98 carries a contact 104 whose position is adjustable by the motion of screw 106. Contact 104 engages a contact 108 on double throw micro-switch 122. When rack 92 is impelled to the left by impact of stop latch 93 on margin stop 90, lever 98 is caused to pivot about pin 100, separating contact 104 from contact 108 and thereby throwing left margin microswitch 122 into the position (NC) in which stepping motor 46 powers a rightward creep of carriage 32 at one quarter of the standard stepping speed. The circuitry which accomplishes this is shown in FIG. 12 and is more fully described in what follows. This sequence of events is shown at time point B in FIG. 9.

Carriage 32 is moved toward the right until the recoil of rack 92 from its left overtravel again closes contacts 104 and 108, which throws left margin microswitch 122 back to its usual position (NO) and thereby stops the rightward creep of carriage 32. By adjustment of screw 106, the final left margin position of carriage 32 can be set with any desired accuracy.

All of the described operations of the modified typewriter, using both standard alpha-numeric spacing and fine increment spacing, may be controlled, if desired, by an external device, such as a calculator, tape, card reader, or the like. In such use, electrical codes are input to the logic circuitry (to be described) of the modified typewriter, where they are decoded to generate the signals that are otherwise generated by the various function and symbol print keys, as described.

Such operation in response to input codes may be performed in two ways, designated "print mode" and "plot (positioning) mode." In print mode operation, characters are printed line by line in a manner similar to conventional operation through keyboard 12. Each symbol print operation is automatically preceded by a horizontal space through a standard alpha-numeric spacing increment, and the fast return may be used. In plot mode, the automatic spacing before each print is inhibited, and input codes are interpreted to determine fine increment spacing through intervals that may vary at each step. An input code of "plot mode" or "print mode" determines whether an automatic space is made with each printing operation.

Referring now to the table of FIG. 5, each signal for printing a symbol or performing a function, normally generated by one of the keys on keyboard 12 and 13, may if desired be generated in automatic operation by decoding a six-bit code. The six bits are designated R1, R2, R2A, R5, T1 and T2. (These designations correspond to the conventional IBM designations of selector latches in the typewriter). The combinations of bit values shown in the first 11 rows of the table are print codes used to generate signals for the printing of the symbols shown. The last five rows contain function command codes used to generate signals for the functions shown. The six bits are first partially decoded (in a manner to be more fully described later) to distin-

guish a print command from a function command. In selecting a print symbol, the six bits are used to operate six solenoids (FIG. 10) which in turn control the mechanical selection of the print symbol by rotating and tilting the type head to the required orientation, in the same manner as selection is accomplished in the unmodified Selectric Input/Output typewriter. Function command codes are used to control the operation of the stepping motors, by means of circuitry including clocks and a counter, in a manner to be more fully described in what follows.

In plot mode, because of the use of stepping motors in the typewriter of this invention to replace standard spacing mechanisms, the symbol selection and spacing operations may be partially overlapped, which makes it possible to plot more rapidly than could otherwise be done. In addition, horizontal and vertical spacing may be carried out simultaneously.

The construction of the standard unmodified Selectric I/O typewriter is such that the spacing operation is initiated mechanically in response to completion of the select/print operation. Therefore, in an unmodified typewriter, neither the type head nor platen is capable of moving during symbol selection, and (except during the return operation) print head and platen can not move simultaneously; rather, these operations are all performed in sequence. In the present embodiment, the symbol selection mechanism and print (symbol impact) mechanism have not been altered, but the carriage and platen drives have been separated, through the introduction of stepping motors, from the selection and print mechanisms. This makes it possible in plotting to perform several operations together and thereby to save substantial amounts of time.

About 55 msec is required for the mechanical process of symbol selection, tilt and rotation of type head, impact and rebound of the head. The paper and type head carriage must remain stationary during the portion of this time during which type head 14 is close enough to the paper that the printed symbol would be smeared by relative motion of paper and carriage. This period occupies about 10 msec of the total printing time. However, during the remainder of the selection time and initial motion of type head 14 towards the paper, the paper and carriage may still be in relative motion. Allowing about 5 msec for variations in the mechanical process of print symbol impact, this means that during about 40 msec of the select/print operation, carriage and platen may be in motion.

In automatic operation, codes for diagonal spacing, namely "step x-, y-," "step x+, y-," "step x-, y+," and "step x+, y+," shown in FIG. 5, may be transmitted, and after decoding, appropriate signals will be generated for controlling stepping motors 46 and 74 for simultaneous carriage spacing and platen rotation. If desired, additional function keys could be provided on auxiliary keyboard 13 to permit manual control of these motions.

A ready/busy feedback signal is sent by the typewriter to the code source. While occupied in moving and printing, the typewriter sends a busy signal; when it is ready to accept another code, the ready signal is sent. Acceptance of new codes and speed of operation thus depend on the time required for each operation.

The spacing operations of the typewriter of the invention, whether determined by use of the function keys on keyboards 12 and 13 or by signals derived from

input codes, including the fast return, and print operations in response to input codes, are all accomplished under the control of the logic circuitry of the typewriter, shown in FIGS. 10 and 11.

In FIGS. 10 and 11, wherein the subscript "0" to a signal designation indicates that the command is input manually through the keyboard, while the subscript "1" indicates a command input as a code, key input section 306 ("Writer Switches") of FIG. 10 includes switches actuated by the function keys shown in FIG. 1, together with additional switches. Print cycle reed switch 120, actuated by magnet 121 carried on the left end of cycle shaft 64 (FIG. 3), provides a mechanically determined indicator of the completion of a print function for initiation of automatic spacing. Double-throw margin microswitch 122 (FIG. 8) is mechanically actuated during the fast return operation as previously described. Carriage switch 124 is normally open, but is closed when carriage return clutch 70 is engaged, in order to de-energize x-stepping motor 46 and permit main motor 66 to power the fast return.

In manual operation, signals from these switches directly determine operation of the timing devices and stepping motors, without the use of codes.

Code inputs, which are not needed for manual operation of the typewriter, are indicated in section 300 (Print/FCN) of FIG. 10, together with a command trigger pulse 320 and a ready/busy feedback signal 322; command pulse 320 causes a stored six bit code to be transferred to the decoder, while ready/busy signal 322 indicates whether the control circuitry is ready to receive and respond to another command, either from the keyboards or from a code source.

"Decode" section 302, also not needed for manual typewriter operation, partially decodes input codes to distinguish print codes from function codes. A print code causes a Print/FCN signal to appear on line 308. Function codes are partially decoded and combined for use in the control circuitry, in the forms of mode signals and various spacing function signals as indicated in section 304.

"Writer solenoid" section 310 contains the six print selection solenoids, with three other solenoids and associated drive units. The print solenoid, again not used in manual operation, powers the mechanism that drives type head 14 against platen 18, in response to the Print/FCN signal on line 308. The Return solenoid is driven in response either to a signal RTN₀, manually input through key 26, or to a signal RTN₁ from decoder 304, and mechanically causes carriage return clutch 70 to be engaged.

Control section 312, shown in greater detail in FIG. 11, receives signals from Writer Switch section 306 or from Print/FCN section 300 and Decode section 304, and controls the operation of stepping motors 46 and 74, as well as of the writer solenoids of section 310.

Referring to FIG. 11, the timing of the spacing functions is controlled by clock pulse generator 330, which sends pulses to the space clock 332, index clock 334, and right creep clock 336. Generator control 331 turns on generator 330 in response to function commands either from the keyboards 12 and 13 through Writer Switch section 306, or from decoder 304. In addition, a reset signal on line 338 turns generator 330 off. Space and index clocks 332 and 334 determine the number of steps taken by the space and index stepping motors.

The same four-bit counter 344 is used for both space and index operations and is enabled by a pulse from either clock; a count of 10 shuts off flip-flop 346 controlling space (x- axis) clock 332, a count of 14 shuts off flip-flop 348 controlling index (y- axis) clock 334. The counter is reset by an appropriate polarity on reset line 338.

When the typewriter operates in response to coded input, each operation is initiated by a command pulse 320, which is input to Print/function section 300 (FIG. 10) to trigger transfer of an assembled code to the decoder. The command pulse 320 also initiates action by the decoder, and together with a print code from decoder 304, actuates the print solenoid (section 310, FIG. 10). Finally, command pulse 320 is used to send a busy signal on Ready/busy line 340 to provide feedback signal 322 (section 300, FIG. 10). A 150 msec busy signal is also provided in response to depression of any of the manual function keys (section 306, FIG. 10); or in response to codes for the spacing functions, shift and return functions.

The 75 msec pulse generated in response to a RTN signal, either manual or coded, is gated at 342 with a PLOT signal to inhibit otherwise automatic indexing after a fast return in plot mode.

In manual operation, print symbols are selected and printed conventionally in response to depression of symbol selection keys on keyboard 12. A spacing function is initiated when depression of a function key on keyboard 12 or keyboard 13 generates an electrical function signal. The signal is input to control section 312; here horizontal and vertical spacing signals are used to start the generation and counting of clock pulses, which in turn determine the number of steps moved by the appropriate stepping motor. The signals further determine the direction of motion. A return signal initiates an automatic index operation through a standard index increment (unless inhibited in PLOT mode) and simultaneously initiates the fast return operation, as previously described.

When operation is in response to stored commands, codes are input to the Decode Section 304, which generates from the function codes electrical signals corresponding to those generated by depression of the function keys; these signals initiate spacing operations as described. Signals generated in response to print codes are used to drive print symbol selection solenoids, with or without automatic horizontal spacing, depending on the mode of operation.

The modified typewriter of the invention is thus operable under manual control as well as in response to stored commands, and can provide either fine increment spacing for the representation of graphical information or standard letter spacing and index spacing for use in typing in the usual manner, as desired.

What is claimed is:

1. An output writer responsive to input signals comprising horizontal space signals and selection and print signals, said writer having

record sheet supporting means,

a type head providing a plurality of alpha-numeric symbols and operable to select and print an alpha-numeric symbol on a record sheet carried on said supporting means,

selection and printing means connected to said type head for moving said type head relative to said carriage means to select a particular alpha-numeric

symbol and sequentially to print the selected alpha-numeric symbol on the record sheet,

selection and printing drive means including a main motor, said selection and printing drive means being responsive to input selection and print signals for operating said selection and printing means,

type head carriage means supporting said type head, said type head carriage means being driven independently of said selection and printing means for independent control and timing of said carriage means and said selection and printing means, and being horizontally movable with respect to said record sheet supporting means,

type head carriage drive means independent of said selection and printing drive means, said type head carriage drive means including a stepping motor and responsive to input horizontal space signals for moving said carriage means horizontally with respect to said record sheet supporting means,

said carriage drive means and said selection and printing drive means being independently responsive to the input horizontal space signals and the selection and print signals respectively for causing said selection and printing means to select and print an alpha-numeric symbol and for causing said carriage to move horizontally with respect to said record sheet.

2. An output writer as claimed in claim 1, wherein said input signals further comprise a select and print completion signal, a plot mode signal and a print mode signal,

said carriage drive means being responsive to said print mode signal in combination with said select and print completion signal to move said type head carriage horizontally with respect to said record sheet through one standard letter space independently of said input horizontal space signals, and said carriage drive means being responsive to said plot mode signal to respond to said input horizontal space signals independently of said select and print completion signal to move said type head carriage horizontally with respect to said record sheet through a distance defined by said horizontal space signals.

3. The output writer of claim 1, wherein said record sheet supporting means is a rotatable platen,

said input signals further comprise vertical space signals, and

said writer further includes platen drive means including a stepping motor for rotating said platen responsive to said input vertical space signals.

4. The output writer of claim 2, wherein said record sheet supporting means is a rotatable platen,

said input signals further comprise vertical space signals and a line return signal, and

said writer further includes platen drive means including a stepping motor for rotating said platen said platen drive means being responsive to said print mode signal in combination with said line return signal to rotate said platen through a standard line space independently of said input vertical space signals, and

said platen drive means being responsive to said plot mode signal to respond to said input vertical space signals independently of said select and print com-

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pletion signal and said line return signal to rotate said platen through a distance defined by said vertical space signals.

5. An output writer responsive to input signals, comprising horizontal space signals, vertical space signals, selection signals and print signals, said writer having a rotatable platen, a type head providing a plurality of alpha-numeric symbols and operable to select and print an alpha-numeric symbol on a record sheet carried on said supporting means, type head carriage means supporting said type head, said type head carriage means being horizontally movable with respect to said record sheet supporting means, selection means connected to said type head for moving said type head relative to said carriage means to select a particular alpha-numeric symbol responsive to an input selection signal, imprinting means connected to said type head for moving said type head relative to said type head carriage means to imprint the selected alpha-numeric symbol on the record sheet responsive to an input print signal, type head carriage drive means independent of said selection means and said imprinting means, said

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type head carriage drive means including a stepping motor for moving said carriage means horizontally with respect to said record sheet supporting means responsive to input horizontal space signals,

platen drive means including a stepping motor for rotating said platen responsive to said input vertical space signals,

timing means responsive to the input horizontal space signals, vertical space signals, selection signals and print signals for producing timing signals, said carriage drive means and said platen drive means being responsive to said timing signals and the input horizontal and vertical space signals simultaneously to move said carriage horizontally with respect to said record sheet and to rotate said platen to move said record sheet vertically with respect to said type head during operation of said writer.

6. The output writer of claim 5, said timing means further being responsive to the selection and print signals for causing said selection means to operate during movement of said carriage and platen drive means.

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